Notes: Antras & Helpman (JPE, 2004)
Global Sourcing

Contents

1 Big picture

- **Question.** How do heterogeneous firms choose among four global sourcing strategies: domestic outsourcing, domestic integration, foreign outsourcing, and foreign integration/FDI?
- **Core mechanism.** Firms need specialized intermediate inputs. Both the final-good producer H and the input supplier M make relationship-specific investments, but contracts are incomplete. Ownership changes ex post bargaining power and therefore ex ante investment incentives.
- **Two decisions are linked.** A firm simultaneously chooses (i) location of input production, North versus low-wage South, and (ii) ownership structure, outsourcing versus vertical integration.
- **Firm heterogeneity.** Final-good producers differ in productivity. Productivity sorting is central: low-productivity firms choose low fixed-cost organizational forms, while high-productivity firms choose high fixed-cost, high-variable-profit organizational forms.
- **Sector heterogeneity.** Sectors differ in headquarter intensity η. Headquarter-intensive sectors put more weight on the final producer's headquarter services, making control rights and integration more valuable.
- **Main sorting result.** In component-intensive sectors, no firm integrates: low-productivity active firms outsource at home, while high-productivity firms outsource abroad. In headquarter-intensive sectors, all four organizational forms can coexist, ordered from low to high productivity as: domestic outsourcing, domestic integration, foreign outsourcing, and foreign integration/FDI.
- **Trade-cost/wage-gap result.** Lower southern wages or lower trade costs make foreign sourcing more attractive. But in headquarter-intensive sectors, this can raise arm's-length outsourcing relative to intrafirm trade because the marginal switch to foreign sourcing often comes from domestic integration into foreign outsourcing.
- **Prevalence result.** Sectors with more productivity dispersion rely more on imported inputs. Within headquarter-intensive sectors, greater dispersion and greater headquarter intensity tend to increase the prevalence of integration.
- **Economic takeaway.** Global sourcing is not just about low wages. It is a joint choice of location and ownership, shaped by productivity heterogeneity, contract incompleteness, bargaining power, headquarter intensity, and fixed versus variable cost trade-offs.

2 Incremental contribution of the paper

- **Combines heterogeneous-firm trade and incomplete-contract theory.** The paper merges Melitz-style within-sector productivity heterogeneity with Grossman-Hart/Antras-style property-rights theory. This makes firm productivity and firm boundaries jointly endogenous.
- **Four-way global sourcing taxonomy.** Earlier work often contrasted exporting versus FDI or outsourcing versus integration separately. This paper studies all four sourcing modes together: domestic outsourcing, domestic integration, foreign outsourcing, and foreign integration.
- **Productivity sorting across organizational forms.** A central innovation is the result that productivity ranks firms across sourcing modes. The least productive active firms choose low fixed-cost domestic outsourcing; the most productive firms choose high fixed-cost foreign integration when the sector is headquarter intensive.
- **Sectoral headquarter intensity as an organizing primitive.** The paper shows how technological characteristics of sectors determine whether integration is useful. Component-intensive sectors favor outsourcing; headquarter-intensive sectors can support integration.
- **Explains intrafirm trade versus arm's-length trade jointly.** The model gives predictions for when international sourcing takes the form of FDI/intrafirm trade versus foreign outsourcing/arm's-length trade.

- **Endogenous prevalence of organizational forms.** The paper derives how industry-level shares of sourcing modes vary with productivity dispersion, headquarter intensity, wage gaps, transport costs, and bargaining parameters.
- **Counterintuitive comparative statics.** Lower southern wages or lower trade costs increase foreign sourcing, but can also increase outsourcing relative to integration. This is not obvious without jointly modeling location and ownership.
- **Empirical roadmap.** Although theoretical, the paper generates cross-industry predictions later used in empirical work on multinational boundaries, intrafirm trade, and the organization of global value chains.

Incremental contribution in one sentence. Antras and Helpman transform global sourcing from a simple low-wage/offshoring decision into a joint firm-boundary and location problem in which productivity heterogeneity determines who outsources, who integrates, who imports, and who engages in FDI.

3 Model objects and measurement

3.1 Countries, sectors, and firms

- There are two countries: North and South. The North has skilled headquarter activities and is the location of all final-good producers.
- The South has lower manufacturing wages and can produce intermediate components at lower variable cost.
- Sectors produce differentiated final goods under monopolistic competition.
- In each sector, final-good producers differ in productivity θ or v.
- Firms learn productivity after entry and then choose an organizational form or exit.

Interpretation: The North-South structure captures the central global sourcing trade-off: cheaper variable production in the South versus higher fixed organizational and contracting costs abroad.

3.2 Inputs and production

Production uses headquarter services h supplied by the final-good producer and manufactured components m supplied by the manufacturing plant operator. Sectoral headquarter intensity is denoted η.

- High η: headquarter services are important; control by the final-good producer is valuable.
- Low η: components are important; incentives for the component supplier are more important.

Interpretation: Headquarter intensity determines whether the organizational problem is mainly about motivating H or motivating M.

3.3 Incomplete contracts

- The final-good producer and supplier cannot write enforceable ex ante contracts on specialized inputs, labor, or revenue.
- After investments are made, the two parties bargain over the surplus.
- Because investments are relationship specific, each party underinvests relative to the first best.
- Ownership changes the outside option in bargaining and therefore shifts ex ante incentives.

Interpretation: The paper's theory of the firm is a property-rights model: ownership matters because it changes ex post bargaining positions and ex ante relationship-specific investment incentives.

3.4 Organizational forms

Let $k \in \{O, V\}$ denote ownership: outsourcing O or vertical integration V. Let $l \in \{N, S\}$ denote the location of component production: North N or South S. The four organizational forms are:

1. O, N: domestic outsourcing;
2. V, N: domestic integration;
3. O, S: foreign outsourcing / arm's-length imports;
4. V, S: foreign integration / FDI and intrafirm trade.

Interpretation: These four forms are the core taxonomy. The paper's key contribution is to explain how firms sort across all four.

3.5 Fixed costs and variable costs

The paper assumes fixed organizational costs are ordered as

$$f_V^S > f_O^S > f_V^N > f_O^N.$$

Southern production has lower variable cost because the southern wage is lower, but foreign sourcing has higher fixed costs. Integration has higher fixed organizational cost than outsourcing, but gives the final-good producer stronger control rights.

Interpretation: Organizational forms differ in both fixed-cost and variable-profit dimensions. High-productivity firms are more willing to pay high fixed costs because they benefit more from high variable profitability.

3.6 Revenue shares and bargaining

Under outsourcing, the final-good producer receives fraction β of revenue in ex post bargaining. Under integration, the final-good producer has a better outside option and receives a larger fraction:

$$\beta_V^N \geq \beta_V^S > \beta_O^N = \beta_O^S = \beta.$$

Interpretation: Integration shifts surplus toward H, encouraging headquarter investment but discouraging component investment by M.

4 Models and theoretical specifications

4.1 Potential revenue

After using demand and production, revenue can be written as a function of productivity, headquarter investment, component investment, and the sectoral demand index:

$$R(i) = Xv \left[\frac{h(i)}{\eta}\right]^{\alpha\eta} \left[\frac{m(i)}{1-\eta}\right]^{\alpha(1-\eta)}.$$

Interpretation: Revenue is multiplicative in productivity and investments. This makes the investment-incentive problem central: whoever receives too little surplus underinvests in the input they control.

4.2 Profit function by organizational form

For each organizational form (k, l), operating profits can be summarized as

$$\pi_k^l(v, X, \eta) = X^{\alpha/(1-\alpha)} v^{\alpha/(1-\alpha)} \psi_k^l(\eta) - w^N f_k^l.$$

- v is firm productivity.
- X is the sectoral consumption index.
- $\psi_k^l(\eta)$ summarizes the variable-profit effect of ownership, wages, and bargaining incentives.
- $w^N f_k^l$ is the fixed organizational cost in northern labor units.

Interpretation: The profit lines in Figures 3 and 4 are linear in transformed productivity. Differences in slopes capture variable-profit advantages; differences in intercepts capture fixed-cost differences.

4.3 Optimal revenue share

If the final-good producer could choose the revenue share freely, it would choose $\beta^*(\eta)$, increasing from zero to one as η rises:

$$\beta^*(0) = 0, \qquad \beta^*(1) = 1, \qquad \beta'(\eta) > 0.$$

Interpretation: As headquarter services become more important, efficiency requires giving more surplus to H. Figure 1 visualizes this benchmark share.

4.4 Firm entry and exit

After drawing productivity, a firm chooses the best organizational form or exits. The exit cutoff v_M solves

$$\max_{k \in \{O,V\}, l \in \{N,S\}} \pi_k^l(v_M, X, \eta) = 0.$$

Free entry requires expected operating profits to cover entry cost:

$$\int_{v_M}^{\infty} \max_{k,l} \pi_k^l(v, X, \eta) dG(v) = w^N f_E.$$

Interpretation: Entry and exit are Melitz-style, but the organizational choice enriches the post-entry sorting pattern.

4.5 Component-intensive benchmark

For low headquarter intensity, $\eta < \eta_M$, the sector is component intensive. Integration is not optimal. Active firms sort as follows:

$$\text{Exit} \quad \rightarrow \quad O, N \quad \rightarrow \quad O, S.$$

Interpretation: When components matter most, giving the supplier strong incentives is more important than giving control rights to the final-good producer. Therefore outsourcing dominates integration.

4.6 Headquarter-intensive benchmark

For high headquarter intensity, $\eta > \eta_H$, all four organizational forms can coexist. Active firms sort as follows:

$$\text{Exit} \quad \to \quad O, N \quad \to \quad V, N \quad \to \quad O, S \quad \to \quad V, S.$$

Interpretation: The most productive firms choose foreign integration because they can afford the highest fixed costs and benefit most from the variable-profit advantage of foreign sourcing and control over headquarter-intensive production.

4.7 Prevalence comparative statics

With a Pareto productivity distribution, the model derives shares of active firms in each organizational form. The main comparative statics are:

- More productivity dispersion raises foreign sourcing.
- In headquarter-intensive sectors, more productivity dispersion raises integration.
- Higher headquarter intensity reduces reliance on imported inputs but raises integration within headquarter-intensive sectors.
- Lower southern wages or lower transport costs raise foreign sourcing and can raise outsourcing relative to integration.
- Greater bargaining power for H makes outsourcing more attractive because it improves H's payoff even without ownership.

Interpretation: The model explains both cross-firm sorting and cross-sector prevalence patterns.

5 Identification and theoretical design logic

5.1 Theoretical identification

This is a theoretical paper, so identification is not empirical. The design identifies mechanisms by isolating the interaction of three primitives:

1. firm productivity heterogeneity;
2. sectoral headquarter intensity;
3. incomplete-contract ownership incentives.

Interpretation: The model's contribution is to show that these three ingredients jointly determine global sourcing patterns.

5.2 Why all three ingredients are needed

- Without productivity heterogeneity, firms in a sector would choose a common organizational form.
- Without incomplete contracts, ownership would not matter for investment incentives.
- Without North-South cost differences, the location decision would not be meaningful.
- Without headquarter intensity differences, the model could not explain why integration is prevalent in some sectors but not others.

Interpretation: The paper's novelty is not one friction alone, but the way several frictions interact to produce rich sorting patterns.

5.3 Main theoretical limitations

- The model is static and does not explicitly model dynamic learning, supplier networks, or repeated relationships.
- All final-good producers are located in the North.
- Only two locations and two ownership modes are considered.
- The paper does not estimate the model empirically.
- The fixed-cost ordering simplifies the taxonomy of cases.

Interpretation: These simplifications make the sorting results clear. Later empirical and quantitative work builds on the framework to test and estimate similar mechanisms.

6 Section-by-section study guide

- **Introduction.** Motivates global production sharing with examples such as Nike, Intel, Barbie, and automobile value chains. Frames the need for a model with endogenous location and ownership.
- **Section II: The Model.** Defines countries, sectors, CES demand, productivity heterogeneity, headquarter services, component inputs, fixed organizational costs, and incomplete contracting.
- **Section III: Equilibrium.** Derives revenue, bargaining payoffs, profit functions, the optimal revenue-share benchmark, and free-entry equilibrium.
- **Section IV: Organizational Forms.** Characterizes sorting patterns for component-intensive and headquarter-intensive sectors. This is the core theoretical section.
- **Section V: Prevalence.** Uses a Pareto productivity distribution to derive how the shares of organizational forms respond to productivity dispersion, headquarter intensity, wage gaps, transport costs, and bargaining power.
- **Conclusion and Appendix.** Summarize the main predictions and show that alternative prevalence measures, such as sales shares, give similar comparative statics.

7 Figures and theoretical result blocks

7.1 Figures 1 and 2: Revenue-share benchmark and organizational-form sorting

Read:

- Figure 1 plots $\beta^*(\eta)$, the revenue share for H that would maximize joint profits if the share could be chosen freely.
- $\beta^*(\eta)$ rises with headquarter intensity because H's investment becomes more important.
- The points β, β_V^S, and β_V^N show the feasible shares generated by outsourcing and integration in each location.
- Figure 2 summarizes the sorting of organizational forms along productivity for component-intensive and headquarter-intensive sectors.

- In component-intensive sectors, active firms only outsource: low productivity at home, high productivity abroad.
- In headquarter-intensive sectors, productivity sorting creates all four forms, ending with the most productive firms integrating in the South.

Detailed interpretation: Figure 1 is the contract-theory core of the paper. It explains why ownership structure matters: the feasible revenue shares created by outsourcing and integration may or may not approximate the efficient revenue share. When the sector is component intensive, lower revenue shares for H preserve incentives for the component supplier. When the sector is headquarter intensive, larger revenue shares for H become efficient. Figure 2 then maps this incentive logic into firm sorting. Productivity determines which fixed-cost/variable-profit trade-off a firm can afford.

Economic message: The first two figures explain the entire architecture: technology determines the desired split of surplus, ownership determines the feasible split of surplus, and productivity determines which ownership-location package a firm chooses.

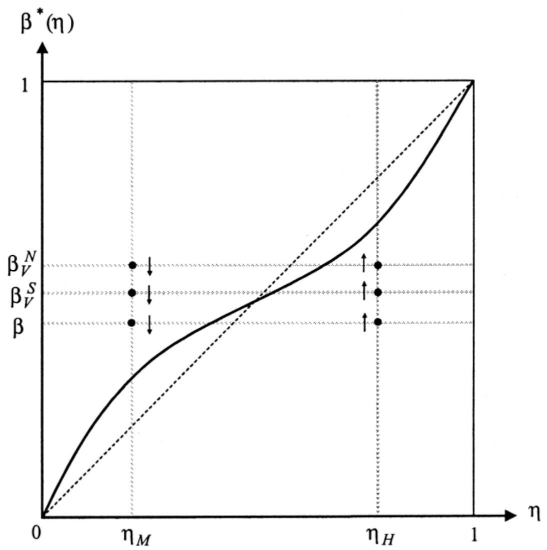


Fig. 1.—Distribution of revenue that maximizes joint profits

(a) Figure 1: Distribution of revenue that maximizes joint profits.

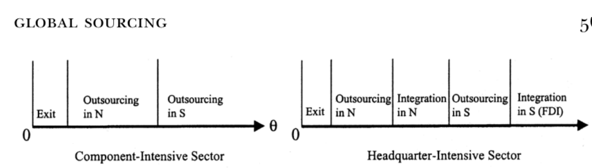


Fig. 2.—Organizational forms

(b) Figure 2: Organizational forms.

7.2 Figures 3 and 4: Equilibrium profit lines by sector type

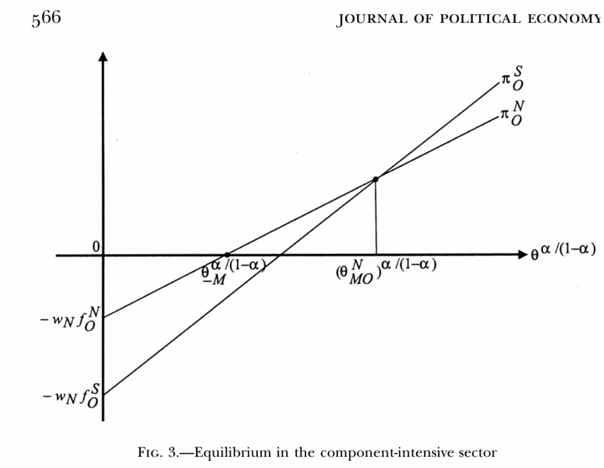
Read:

- Figure 3 shows the component-intensive sector. Only two profit lines matter: domestic outsourcing and foreign outsourcing.
- The steeper foreign outsourcing line reflects lower southern variable costs, while the lower intercept reflects higher foreign fixed costs.
- Low-productivity firms exit, intermediate-productivity firms outsource in the North, and high-productivity firms outsource in the South.
- Figure 4 shows the headquarter-intensive sector. Four profit lines can matter: O, N, V, N, O, S, and V, S.

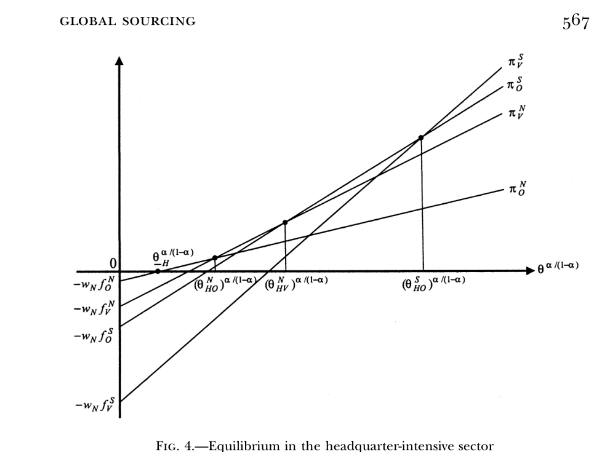
- The high-productivity ranking becomes domestic outsourcing, domestic integration, foreign outsourcing, and foreign integration/FDI.

Detailed interpretation: Figures 3 and 4 are the central equilibrium diagrams. They show how the same two economic forces - fixed costs and variable profits - create productivity cutoffs. The slopes of the lines encode variable-profit advantages; the intercepts encode fixed organizational costs. A more productive firm gains more from a higher slope, so it is willing to pay higher fixed costs. Figure 4 is richer because integration becomes valuable when headquarter services are important.

Economic message: The organizational boundary of the firm is endogenous. The most productive firms do not simply export more; they also choose more complex ownership structures and locations.



(a) Figure 3: Equilibrium in the component-intensive sector.



(b) Figure 4: Equilibrium in the headquarter-intensive sector.

7.3 Theoretical result block 1: productivity sorting

- Low-productivity firms choose low fixed-cost organizational forms or exit.
- High-productivity firms choose high fixed-cost, high-slope forms.
- Foreign sourcing is chosen by sufficiently productive firms because the southern wage advantage creates higher variable profits.
- Integration is chosen by sufficiently productive firms in headquarter-intensive sectors because control rights raise H's investment incentives.

Interpretation: Productivity is the sorting variable that converts organizational trade-offs into observable firm-level heterogeneity.

7.4 Theoretical result block 2: sectoral headquarter intensity

- Component-intensive sectors favor outsourcing because supplier incentives are central.
- Headquarter-intensive sectors favor integration because H's investment is central.
- Higher headquarter intensity reduces the importance of low-cost component manufacturing and therefore can reduce imported-input prevalence.
- Within headquarter-intensive sectors, higher headquarter intensity raises integration.

Interpretation: This explains why global sourcing patterns vary by industry, not only by firm productivity.

7.5 Theoretical result block 3: productivity dispersion

- More productivity dispersion raises the mass of high-productivity firms.
- In component-intensive sectors, this raises foreign outsourcing.
- In headquarter-intensive sectors, this raises imported-input use and also raises integration relative to outsourcing.
- This mirrors Melitz-style reallocation, but with organizational forms replacing simple export participation.

Interpretation: The model turns productivity dispersion into a determinant of intrafirm trade and arm's-length trade.

7.6 Theoretical result block 4: wage gaps, transport costs, and bargaining power

- Lower southern wages or lower transport costs raise foreign sourcing.
- In headquarter-intensive sectors, these changes can shift firms from domestic integration into foreign outsourcing, increasing arm's-length trade relative to FDI.
- Greater primitive bargaining power for H makes outsourcing more attractive because H can obtain more surplus without ownership.

Interpretation: These comparative statics show that globalization can increase outsourcing even when multinational firms also grow.

8 Result map

- **Input-intensity result:** headquarter-intensive sectors are more likely to integrate; component-intensive sectors are more likely to outsource.
- **Productivity result:** high-productivity firms choose foreign sourcing and, in headquarter-intensive sectors, foreign integration.
- **Wage-gap result:** lower southern wages expand foreign sourcing and can increase outsourcing relative to integration.
- **Dispersion result:** greater firm productivity dispersion increases imported-input use and, in headquarter-intensive sectors, integration.
- **Bargaining-power result:** stronger bargaining power for H makes outsourcing more attractive relative to integration.
- **Empirical implication:** intrafirm trade shares should be higher in headquarter-intensive sectors, especially when productivity dispersion is large.

9 Short summary

- Antras and Helpman develop a North-South model of global sourcing with heterogeneous firms and incomplete contracts.
- Final-good producers are located in the North and choose where and how to source specialized components.
- There are four organizational forms: domestic outsourcing, domestic integration, foreign outsourcing, and foreign integration/FDI.
- Ownership affects bargaining and investment incentives.
- Location affects variable costs and fixed organizational costs.
- Productivity sorts firms across sourcing modes.
- Component-intensive sectors feature outsourcing only.
- Headquarter-intensive sectors can feature all four sourcing modes.
- More productive firms source abroad; among firms sourcing in the same country, more productive firms integrate.
- Lower southern wages and lower trade costs increase foreign sourcing but can also increase outsourcing relative to integration.
- Productivity dispersion and headquarter intensity determine the prevalence of intrafirm versus arm's-length trade.

10 Conclusion

10.1 Core conclusion

Antras and Helpman show that the international organization of production is shaped jointly by firm productivity, sector technology, wage differences, fixed organizational costs, and incomplete contracts. Global sourcing is not a binary offshoring decision; it is a four-way choice over ownership and location.

10.2 Economic intuition

Foreign production offers lower variable costs but higher fixed organizational costs. Integration gives the final-good producer stronger control rights but can weaken supplier incentives and raises fixed costs. More productive firms can afford higher fixed costs, so they choose organizational forms with higher variable profitability. Sectors with high headquarter intensity make integration more attractive because the final-good producer's investment is especially important.

10.3 Incremental contribution revisited

The paper's lasting contribution is to combine heterogeneous-firm trade theory with property-rights theory of the firm. This combination explains why different firms in the same industry choose different global sourcing modes and why the prevalence of intrafirm trade and outsourcing varies systematically across industries.

10.4 Implications for empirical work

The model predicts that empirical studies should examine not only whether firms import inputs, but also whether those inputs are sourced from affiliated firms or independent suppliers. It also predicts that industry headquarter intensity and firm productivity dispersion should explain cross-industry variation in intrafirm trade shares.

10.5 Limitations

The model is theoretical and stylized. It abstracts from dynamic supplier relationships, multi-country sourcing networks, financial constraints, and empirical estimation. It also assumes all final producers are in the North and restricts attention to two ownership modes and two locations. These simplifications make the mechanism clear but leave room for empirical and quantitative extensions.

10.6 Takeaway

The key takeaway is that globalization changes not only where firms produce, but also how they organize production. Trade, FDI, outsourcing, and firm boundaries are jointly determined outcomes of productivity, technology, and contracts.